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Employees' perceptions of the fairness of AI-based performance prediction features

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ABSTRACT
Predictive Artificial Intelligence (AI) algorithms are being increasingly adopted by organizations to inform workforce-related decisions. However, their effectiveness could be undermined by a lack of ground-truth data, particularly concerning human perceptions of fairness in predictive factors. This study examined 306 employees' perspectives on the fairness of 31 predictive features in five categories: demographic, academic, work-related, cognitive abilities and personality traits, and professional development and certification. Our findings revealed that employees generally regard features as fair when they align closely with job performance and contribute to prediction accuracy. Features related to technology usage attitude, workplace comfort, job rank satisfaction, alignment between one's field of expertise and work, years of service in the same field, and total work experience were the highest perceived fair predictors. Conversely, more participants tended to perceive features such as gender, marital status, and number of children as unfair due to concerns about their actual contribution to prediction accuracy, privacy implications, and the risk of bias and discrimination. Features pertaining to professional development and certification were considerably viewed as fair predictors. These findings are valuable for the development of fairer AI algorithms for predicting employee performance by ensuring that predictive features align with employee perceptions of fairness.

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1. Introduction

Employee performance reflects an individual's accomplishments and effectiveness in their roles or occupations. It encompasses the outcomes achieved by employees in terms of the quality and quantity of their work, aligning with their assigned responsibilities (Aliyyah et al., 2021; Shahzadi et al., 2014). The success of an organization depends on the contributions of its workforce, and the effectiveness of individual employees plays a determining role in the overall success of the organization (Brefo-Manuh et al., 2016). Employee performance reinforces the organization's productivity and output and it is the path to the organization's achievements (Brefo-Manuh et al., 2016). Consequently, the assessment of employee performance has become an essential practice and a tool for enhancing productivity in modern organizations (Obi, 2016).

Many organizations maintain a structured performance appraisal system, where employee job performance is routinely assessed, typically on an annual or semi-annual basis (Begum et al., 2015; Carroll & Schneier, 1982; Walsh, 2003). A well-designed performance evaluation system has the potential to enhance employees' performance and organization's operations (Brefo-Manuh et al., 2016; Vanci-Osam & Aksit, 2000). It serves a dual purpose by providing employees with clear expectations and generating vital data for making employment-related decisions, including considerations related to salary increments, promotions, or terminations (Brefo-Manuh et al., 2016; Lillian et al., 2011; Locher & Teel, 1988; Rao, 2009;

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Walsh, 2003). However, the assessment of employee performance has traditionally depended on conventional techniques and technologies.

In the modern practice of Human Resource Management (HRM), there has been a notable integration of Artificial Intelligence (AI), with leading technology firms, such as Amazon (Black & van Esch, 2020; Dastin, 2018, 2022) and IBM (Black & van Esch, 2020; Guenole & Feinzig, 2018), incorporating algorithmic processes for various HR functions. According to research conducted in 2017, 40 percent of the human resource functions of international companies use AI-based tools (PWC, 2017). Artificial intelligence is taking place in the context of employee performance assessment, revolutionizing the approach to employee performance assessment (Mahmoud et al., 2019; Park et al., 2022). In recent years, machine learning and algorithm models have been increasingly used to analyze and predict the performance of employees in organizations (e.g., Jayadi et al. (2019); Patel et al. (2022)). While innovative, this trend has not been without challenges and debates. Studies have begun to highlight the attitudes, perceptions, and expectations of employees towards AI in HRM, such as the perceived fairness of AI-based job performance evaluations (Park et al., 2021, 2022). This growing body of research on employees' perceptions of AI in HRM can be viewed as an extension of traditional studies on perceived fairness in HR practices, such as perceived fairness of employee performance appraisal (e.g., Walsh (2003)), and as an integral component of ongoing research efforts focused on ethical and fair AI (e.g., Park et al. (2022)).

Understanding how humans perceive fairness is paramount in the pursuit of human-centered AI (Lee et al., 2020; Riedl, 2019; Shulner-Tal et al., 2023). This understanding plays a central role in guiding the actions of designers and developers who bear the responsibility for developing AI algorithms and systems that meet ethical standards (Kieslich et al., 2022; Xivuri & Twinomurinzi, 2023). By gaining insight into how individuals interpret fairness, informed decisions can be made during the design and development process of algorithms, ensuring that AI systems align with ethical and societal expectations (Shulner-Tal et al., 2023). The significance of comprehending human perceptions of algorithmic fairness also extends beyond AI-based solution development. It provides valuable insights for decision-makers tasked with evaluating the feasibility and appropriateness of implementing AI-based systems in diverse organizational and societal contexts (Kieslich et al., 2022). The consideration of human perceptions would inform the strategic choices surrounding the adoption of AI technologies, allowing decision makers to assess potential benefits, ethical implications, and alignment with societal values (Cao et al., 2021; Whittlestone et al., 2019).

Predicting employee performance has emerged as a prominent application of predictive algorithms and algorithmic decision-making (ADM), empowering organizations with the capability to make decisions regarding their workforce (Patel et al., 2022). It is important to clarify at this stage that some studies distinguish between predictions and decisions (Beigang, 2022; Corbett-Davies & Goel, 2018; Hedden, 2021; Kleinberg et al., 2018). A predictive algorithm may be highly fair, but its predictions can be used to produce unfair decisions (Hedden, 2021). Therefore, predictive algorithms are oriented towards generating specific predictions, whereas decision-making algorithms use these predictions, with or without other criteria, to make choices, decisions, or recommendations. These two often work together in various applications to support better decision-making processes (Beigang, 2022).

AI-based predictive algorithms employ a set of features to increase prediction accuracy. Humans can perceive these features as fair or unfair. In the context of employee performance prediction, algorithms are developed by considering several factors, such as individual characteristics, prior experience, level of schooling, and socioeconomic status (e.g., Mahmoud et al. (2019); Patel et al. (2022)). However, some of these factors can be perceived as unfair by individuals or specific groups in diverse workplace scenarios. Previous research has emphasized the necessity of developing fairness-aware or fair algorithms (Park et al., 2022). Nonetheless, the limitation of ground-truth data, specifically the understanding of how humans perceive the fairness of predictors and their integration into algorithmic decision-making, may hinder the efficacy or success of developing fair algorithms.

Our research holds significant relevance in the context of fairness perception, a notion inherently dependent on the specific context in which it is applied. The concept of algorithmic fairness is not universally uniform, and its importance and challenges differ across domains. In the domain of HR and employee performance prediction, this study assumes paramount importance because of its inherent complexity. First, employee performance is influenced by a multitude of factors, necessitating the

consideration of several variables in the prediction algorithm. A critical question arises: Which of these variables are perceived as fair or unfair by the employees themselves? Second, the consequences of algorithmic decisions in this domain extend to multiple levels. At the individual level, the effects may involve job loss, salary disparities, diminished job satisfaction, and morale, along with the potential initiation or perpetuation of bias and discrimination. At the organizational level, effects may manifest as talent drain, increased employee turnover, reduced diversity, lower productivity and innovation, reputation damage, legal liabilities, and ethical dilemmas. At the societal level, these effects may contribute to broader inequalities.

The primary objective of this study is to explore employees' perceptions of fairness regarding various predictors used in AI-based algorithms for predicting employee performance. By uncovering these perceptions, this study aims to inform and guide the development of fairer AI algorithms for performance assessments. This study also investigated the underlying reasons for employees' perceptions of fairness for each predictive feature. Exploring the motivations and rationale behind these perceptions would provide a better understanding of perceived fairness, representing an advancement in this area of research. The study also examined the correlations between these perceptions, offering a comprehensive and interconnected view of how employees perceive the fairness of performance predictive features in AI algorithms.

Overall, this study aims to address three fundamental research questions:

RQ1: How do employees perceive the fairness of features used in performance prediction algorithms?

RQ2: What are the underlying factors and rationales influencing employees' perceptions of the fairness or unfairness of performance-predictive features?

RQ3: To what extent do employees' perceptions of the fairness of predictive features correlate?

This paper is divided into five sections. Section 2 provides background information and discusses related studies. The methods used in this study are described in Section 3. Section 4 presents and discusses our findings. In Section 5, we summarize our main results, acknowledge the limitations of the research, and suggest ideas for future research.

2. Background and related work

2.1. The concept of fairness

Fairness is a central element in the design and development of trustworthy AI systems that, aiming to mitigate undesirable outcomes (Hutchinson & Mitchell, 2019). It is a key focus in over 80% of AI ethics guidelines, as highlighted by Jobin et al. (2019). The Organization for Economic Co-operation and Development (OECD) lists it as one of the core principles for AI (OECD, 2019). It is also the foremost principle in the Saudi Authority for Data and Artificial Intelligence (SDAIA) AI ethics principles (SDAIA, 2023). The European Commission further recognizes it as one of the seven essential requirements for AI systems to be deemed trustworthy (European Commission, 2019).

The concept of "fairness" in AI ethics emphasizes the necessity of ensuring equitable outcomes. This principle dictates that algorithmic decisions should not lead to unjust, discriminatory, or disproportionate consequences (Shin & Park, 2019). The literature presents various descriptions and explanations of "fairness." Verma and Rubin (2018) conducted a comprehensive review, revealing that formal fairness concepts can be broadly categorized into three groups: statistical measures, similarity-based measures, and causal reasoning. Statistical measures involve calibrations of predicted probabilities, predicted outcomes, and actual outcomes. Similarity-based measures advocate for treating similar individuals similarly irrespective of their group classification. Causal reasoning suggests using structural equations to estimate the impact of sensitive attributes and to design algorithms that ensure acceptable discrimination levels by taking these attributes into account.

Other studies explored AI fairness within four dimensions: distributive, procedural, informational, and interpersonal (Starke et al., 2022). Distributive fairness involves the equitable allocation of resources based on principles of equality, equity, or need (Deutsch, 1975). Procedural fairness requires prediction

or decision-making based on fair and just criteria (Levanthal, 1980). In the context of procedural fairness, input features used in the decision process are considered, evaluated, and morally judged by humans regarding the fairness of their use (Grgić-Hlača et al., 2018). However, while certain studies approach fairness with an emphasis on the inclusion of sensitive or protected input features, others consider it from a wider perspective, accounting for various dimensions of a process (Hsu et al., 2021; Marcinkowski et al., 2020; Schlicker et al., 2021; Starke et al., 2022). Informational fairness focuses on the transparency of an algorithmic system (Greenberg & Cropanzano, 1993). Interpersonal fairness involves that algorithmic system respects privacy rights and avoids using protected data (Greenberg & Cropanzano, 1993). Distributive and procedural are the two primary categories of fairness concerns, as per the organizational justice literature (Greenberg, 1987). However, it is argued that the existing body of AI fairness literature predominantly directs its attention towards distributive fairness, with only a few studies on procedural fairness (Grgić-Hlača et al., 2018; Robert et al., 2020).

2.2. Employee performance prediction

To predict employee job performance, a range of predictor variables, commonly referred to as features, are essential (Lather et al., 2019; Patel et al., 2022; Pathak et al., 2023; Thakur et al., 2015). These features span various categories, including demographic details like age, gender, marital status, and number of children; academic qualifications such as scientific specialization and education degree; work-related aspects including job role, level or rank, and title, the number of organizations worked for, previous performance ratings, total years of work experience, tenure in the current organization, perceptions of the work environment, and satisfaction with salary and job rank; as well as training and professional development components (Mahmoud et al., 2019; Patel et al., 2022). The selection of these features is grounded in a substantial body of literature that investigates the diverse factors that influence employee performance (Diamantidis & Chatzoglou, 2018; Elnaga & Imran, 2013).

The relevance of performance prediction features can vary across different work sectors, domains, or scenarios. Some features may be universally applicable (Anyim et al., 2012; Rahaman et al., 2021), while others are more pertinent to specific contexts (Patel et al., 2022; Tinofirei, 2011). Selbst et al. (2019) emphasized that algorithmic solutions developed for one social setting, such as a public sector organization, may not be suitable or effective in different contexts, such as the private sector. This was confirmed by Hunkenschroer and Luetge (2022), who noted that perceptions of fairness can differ significantly across various contexts. Furthermore, certain features might be more predictive, depending on the employee's level of experience. Tracey et al. (2007) found that general mental ability is a more effective indicator for new employees, whereas conscientiousness is more predictive for those with more experience.

It is important to note that developing fair algorithmic systems necessitates careful selection of features, especially those representing sensitive or protected attributes (Morse et al., 2021; Verma & Rubin, 2018). Sensitive attributes include demographic characteristics, such as gender, age, disability status, marital status, ethnicity, nationality, and socioeconomic status. As a fundamental principle, features associated with sensitive or protected attributes should be avoided as predictors in machine-learning models (Morse et al., 2021). While some existing machine learning models incorporate sensitive attributes to predict employee performance, our study aims to understand how employees perceive the fairness of using these features in performance predictions.

2.3. Related work

Research on fair AI is primarily divided into two distinct areas (Park et al., 2022). The first focuses on the technical aspects of fairness (i.e., algorithmic fairness), addressing issues such as algorithmic bias and discrimination, which typically arise when algorithms unintentionally favor or disadvantage specific individuals or groups based on attributes such as gender, race, or ethnicity. Studies in this domain include efforts to detect and correct bias in training datasets, defining "fairness" mathematically for model optimization, and developing equitable algorithms by incorporating fairness-centric mathematical functions (Jayadi et al., 2019; Leung et al., 2020; Patel et al., 2022; Selbst et al., 2019).

In employee performance prediction, several studies have utilized machine learning for algorithmic evaluations. For instance, Jayadi et al. (2019) implemented machine learning and the Cross-Industry Standard Process for Data Mining (CRISP-DM) to predict employee performance using a dataset of 310 employee records with 28 variables (e.g., gender, age, pay rate, marital status, employee status, performance score, position, and termination status), achieving a classification accuracy of 95.48% with the Naive Bayes method. Patel et al. (2022) analyzed employee performance and provided ratings using factors from the professional and personal spheres of employees. Their research compared multiple AI-based algorithms, including Random Forest, Artificial Neural Network, Decision Tree, and XGBoost, and introduced an ensemble method named "RanKer." Utilizing a dataset of 1,200 employees with 28 variables (e.g., age, gender, marital status, job role, education level, job level, job satisfaction, work history, work-life balance, department, attrition, and performance rating), the RanKer model demonstrated superior performance over traditional models in terms of precision, recall, F1-score, and accuracy.

However, these technical studies may not fully address concerns about the fairness of performance prediction algorithms, as they often overlook human perceptions of system fairness (Hunkenschroer & Luetge, 2022), which are crucial for the system's real-world success. Our study aims to fill this gap by exploring employee perspectives on the fairness of features used by machine-learning models for employee performance prediction. The insights gained are intended to inform and inspire the design of fairer algorithms, and enrich the discussion on the fairness of AI-based predictions of job performance.

The second area of research focuses on achieving socially fair outcomes, emphasizing fairness as perceived by individuals or groups rather than solely on the quantifiable and technical aspects of algorithmic fairness (Park et al., 2022). However, research in this domain is less prevalent (Robert et al., 2020). One of the studies in this area was conducted by Lee (2018), who found that employees often view AI-based evaluations doubtfully, especially when compared to human-led assessments, questioning their fairness and trustworthiness. In contrast, a recent study by Qin et al. (2023, p. 1) discovered that "employees consider AI to be fairer and more accurate than average human managers in evaluating their performance."

Furthering this line of research, Park et al. (2021) investigated the factors influencing employee perceptions of unfairness in algorithmic management, with a specific focus on work performance evaluations. They identified a range of challenges including emotional and cognitive influences, potential bias, manipulation concerns, and privacy violations. In a later study, Park et al. (2022) focused on conflicts that emerge from the use of AI in HRM, involving various stakeholders, such as employees. Their study aimed to identify design solutions to address these challenges. A key finding from their study was related to "fairness," highlighting that employees expect objective evaluations in work performance assessments. They reached a consensus that achieving fairness is possible by establishing transparent and equitable processes, thereby ensuring procedural fairness in algorithmic decision-making.

Existing research has enhanced our understanding of employee perspectives on algorithmic decision-making in HRM, particularly concerning work performance evaluation. However, our research aimed to address certain limitations in this area. First, while previous studies have yielded valuable qualitative insights into general human perceptions of AI in HRM and performance evaluations, our study is specifically designed to deepen the understanding of employee perspectives on the fairness of specific AI features, including exploring the reasons behind these perceptions. Second, we intend to augment the qualitative data from prior research with a quantitative analysis, providing a more comprehensive view of the subject matter. Third, there is an evident gap in research examining employee perceptions across diverse social, organizational, and cultural contexts, as these factors influence notions of fairness (Hunkenschroer & Luetge, 2022). Accordingly, our study specifically focused on exploring the perceptions of employees within the unique organizational and cultural context of Saudi Arabia, examining their views on the fairness of algorithmic features used in predicting employee performance.

3. Methodology

In this study, we employed a survey methodology. An online questionnaire was developed and distributed from September 2021 to February 2022 to collect employees' perceptions regarding the fairness of

performance predictive features. This section describes the design and validation of the questionnaire, details the composition of the sample, and addresses the ethical considerations.

3.1. Questionnaire design

The questionnaire was comprised of three main sections. The first section involved demographic questions, aimed at gathering data on gender, age, education level or degree, scientific specialization, job type (supervisory or non-supervisory), type of organization, years of work experience in the current organization, and nationality.

The second section contained a single question assessing the respondents' understanding of the basic working principles of artificial intelligence systems (intelligent algorithms). Respondents rated their understanding on a 7-point Likert scale (1 = very weak, 7 = very advanced). This question aimed to ensure a diverse sample in terms of understanding algorithms, reflecting the real-world scenario in which employees have varying levels of familiarity with AI systems. Unlike Shin (2020), who used understanding as a screening tool, our goal was to capture a broad range of perceptions, regardless of the AI knowledge level.

The third section listed 31 predictive features of employee performance, drawn from AI-based performance prediction models (e.g., Mahmoud et al. (2019); Patel et al. (2022)) and literature on factors affecting employee performance (e.g., Diamantidis and Chatzoglou (2018); Elnaga and Imran (2013)). For the list of predictive features, refer to Table 2. Participants were asked to rate their agreement on a 5-point Likert scale (1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree) on the fairness of using each feature in intelligent algorithms for performance prediction. They also identified motivations for their ratings, choosing from six multi-select options derived from the literature (e.g., Diamantidis and Chatzoglou (2018); Elnaga and Imran (2013), Afrin et al. (2022)). These motivations included: increasing prediction accuracy, not enhancing prediction accuracy, collecting sensitive data or violating privacy, causing discriminatory decisions, causing biased decisions, or other motives. It is important to mention that discriminatory and biased decisions, while related, differ in some aspects. Discriminatory decisions pertain to differential treatment based on characteristics like race or gender, which may be intentional or unintentional (Prince & Schwarcz, 2019; Takshi, 2020). Biased decisions refer to systematic distortions in decision-making, often stemming from factors such as biased data, flawed algorithms, or even human bias in the design or implementation of the system. Nonetheless, while bias plays a crucial role in discriminatory processes, it does not always lead to discrimination (Ferrer et al., 2021).

3.2. Questionnaire validation

To ensure content validity, the questionnaire underwent a rigorous pre-testing process as suggested by Zikmund et al. (2012). Initially, it was reviewed by three computer science experts who are knowledgeable in intelligent algorithms. They critically assessed various elements of the questionnaire, including the clarity of the instructions and questions, the appropriateness of the question sequence, and the relevance of the predictive features included. Based on their feedback, some modifications were made to enhance the clarity. Notably, there was a consensus among all experts regarding the proposed features.

A cognitive test was also conducted with an AI specialist. During this test, the specialist articulated their thought process while completing the questionnaire, allowing us to identify and address issues such as ambiguous expressions or the use of overly technical terms (Draugalis et al., 2008; Willis, 2004). The researcher also proactively inquired about potential problems, especially when hesitations or uncertainties were observed during the specialist's think-aloud process (Collins, 2003).

Following these expert reviews, a pilot version of the questionnaire was posted on Twitter and LinkedIn. The post included a clear introduction and explanation of the study's purpose, inviting responses from employees in Saudi Arabia, our target population. Analysis of these initial responses was important for detecting any inconsistencies or unexpected patterns, enabling us to further refine the questionnaire, such as clarifying terminology and adjusting the question format for fairness ratings and their motivational underpinnings.

With these adjustments and having achieved a satisfactory level of validity, we proceeded to collect data from the final sample for our study.

3.3. The sample

The sample for this study was collected through collaboration with five trainers who distributed an online questionnaire among employees attending various training programs at a government institute in Saudi Arabia. The institute conducts training sessions for nearly 4,000 employees weekly. Throughout the data collection period, approximately 1,000 employees across forty training programs were invited to participate. The programs were carefully selected to represent three competency levels: novice, intermediate, and proficient, spanning diverse training sectors such as human resources, business, public administration, and information technology. To mitigate potential biases in data collection, the characteristics of employees attending these targeted programs were not reviewed prior to their selection. Out of the total invited employees, 356 accepted the invitation, resulting in a response rate of 35.6%. Ultimately, 306 participants completed all required questionnaire items, yielding a completion rate of 30.06%.

All participants were Saudi nationals working across various sectors in Saudi Arabia, including education, health, finance, commerce, energy, industry, transportation, information technology and communication, municipal services, justice, religious affairs, internal and external affairs, military, bureaus, human resources, and social services. The organizations represented by the participants were predominantly governmental (66.67%), with some from semi-governmental (9.80%) and private sectors (23.53%). For more demographic information, refer to Table 1.

In terms of participants' understanding of AI, respondents demonstrated an above-average familiarity with the basic working principles of intelligent algorithms, as indicated by a mean score of 4.0 and a standard deviation of 1.26 on a 7-point Likert scale.

3.4. Ethical considerations

The ethical approval for this study was exempted as it did not involve the collection of sensitive information from employees. Informed consent was collected electronically. Prior to accessing the survey questions, participants were presented with detailed information about the research, including its objectives and procedures. Consent was obtained through a digital form integrated into the questionnaire. Participants had to explicitly agree to participate, indicating their understanding of the project and their voluntary consent to engage in the research. They were also informed of their right to withdraw at any stage during the survey. Participants were assured that their anonymized data would be used exclusively for research purposes.

Table 1. Profile of the participants.[a]

Demographic category	Option	Percentage
Gender	Male	64.71%
	Female	35.29%
Age	21–29	19.61%
	30–39	56.86%
	40–49	17.65%
	50–59	1.96%
	>=60	3.92%
Education	High school or equivalent	5.88%
	Bachelor's degree	49.02%
	Master's degree	17.65%
	Doctoral Degree	9.80%
	Other: diploma	17.65%
Specialization	IT related	17.65%
	Non-IT related	82.35%
Job type	Supervisory role	33.33%
	Non-supervisory role	66.67%
Years of experience	<2	7.84%
	2-4 years	29.41%
	5-9 years	23.53%
	>=10	39.22%

[a]Sample, $n = 306$.

4. Results and discussions

4.1. Overall results of perceptions and motivational underpinnings

Participants' views on the fairness of the various predictive features used for performance prediction showed notable diversity. Table 2 offers a clear indication of the general sentiment towards each feature by detailing the total disagreement (combined percentages of 'strongly disagree' and 'disagree') and total agreement (combined percentages of 'agree' and 'strongly agree'). Complementarily, Table 3 presents the underlying motivations that shaped these perceptions. It shows the proportions for reasons behind employees' judgments of certain features as fair or unfair in the context of performance prediction. In this table, total positive motivations correspond to the results of a single motivation – 'will increase prediction accuracy,' while total negative motivations encompass the combined results of all other motivations, excluding 'other motives,' which was neither viewed as positive nor negative motivation.

Overall, technology usage attitude (TechUseAtt), work environment comfort (WorkEnvComfort), rank satisfaction (RankSat), field specialization match (FieldSpecMatch), years of service in the same field (YrsFieldServ), and total work experience (TotWorkExp) features received the highest agreement rates (ranging from 80% to 88%) as fair predictors of job performance. Conversely, gender, marital status (MarStat), number of children (NoChild), and income rate (IncRate) received the lowest levels of agreement, at 35%, 31%, 22%, and 35%, respectively, indicating possible concerns about their fairness in performance prediction.

There is a clear pattern indicating that employees perceive features as fair predictors when they relate directly to job performance and augment prediction accuracy. For instance, work environment comfort (WorkEnvComfort) emerged as a fair predictor. Employees perceived it as an important contributor to the accuracy of performance prediction, likely because a comfortable work environment is believed to positively influence employee productivity and, thereby, impact job performance. Similarly, field specialization match (FieldSpecMatch) is highly regarded as a fair predictor and a substantial contributor to prediction accuracy. This feature is likely seen as directly influencing job performance, especially in roles where alignment between one's specialization and job requirements is crucial.

Table 2. Proportions of employees' perceptions of fairness of performance predictive features.[a]

Dimension	Feature	Strongly disagree	Disagree	Neutral	Agree	Strongly agree	Total disagreement	Total agreement
Demographic	Age	0.08	0.18	0.20	0.47	0.08	0.25	0.55
	Gender	0.16	0.35	0.14	0.24	0.12	0.51	0.35
	MarStat	0.18	0.35	0.16	0.22	0.10	0.53	0.31
	NoChild	0.20	0.37	0.22	0.14	0.08	0.57	0.22
	IncRate	0.04	0.33	0.27	0.25	0.10	0.37	0.35
Academic	SciSpec	0.02	0.18	0.18	0.33	0.29	0.20	0.63
	SciDeg	0.02	0.16	0.12	0.43	0.27	0.18	0.71
	GPA	0.08	0.20	0.25	0.37	0.10	0.27	0.47
	EduInsEval	0.08	0.31	0.14	0.37	0.10	0.39	0.47
	DegCountry	0.06	0.25	0.24	0.29	0.16	0.31	0.45
Work-related	PrevPerf	0.02	0.08	0.24	0.55	0.12	0.10	0.67
	YrsOrgServ	0.02	0.10	0.16	0.53	0.20	0.12	0.73
	YrsFieldServ	0.02	0.08	0.10	0.49	0.31	0.10	0.80
	NoOrgsWorked	0.04	0.18	0.12	0.49	0.18	0.22	0.67
	TotWorkExp	0.00	0.10	0.10	0.55	0.25	0.10	0.80
	JobTitle	0.02	0.22	0.25	0.37	0.14	0.24	0.51
	JobRank	0.02	0.20	0.22	0.43	0.14	0.22	0.57
	WorkEnvComfort	0.02	0.04	0.08	0.53	0.33	0.06	0.86
	SalarySat	0.00	0.08	0.16	0.35	0.41	0.08	0.76
	RankSat	0.00	0.02	0.12	0.45	0.41	0.02	0.86
	FieldSpecMatch	0.00	0.04	0.12	0.47	0.37	0.04	0.84
	LangOrgMatch	0.00	0.12	0.16	0.43	0.29	0.12	0.73
	TechUseAtt	0.00	0.02	0.10	0.53	0.35	0.02	0.88
Cognitive abilities and personality traits	NumCogTest	0.06	0.20	0.22	0.39	0.14	0.25	0.53
	VerbCogTest	0.10	0.22	0.27	0.27	0.14	0.31	0.41
	LogReasonTest	0.06	0.22	0.29	0.29	0.14	0.27	0.43
	PersTraitTest	0.08	0.12	0.29	0.37	0.14	0.20	0.51
Professional development and certification	TrainHrs	0.02	0.04	0.24	0.51	0.20	0.06	0.71
	WebTrainTime	0.02	0.04	0.25	0.55	0.14	0.06	0.69
	ProfCerts	0.04	0.08	0.16	0.49	0.24	0.12	0.73
	CertBodyEval	0.02	0.04	0.22	0.47	0.25	0.06	0.73

[a]Sample, $n = 306$.

Table 3. Proportions of motivations of fairness perceptions of employee's performance predictive features.[a]

Dimension	Feature	Will increase prediction accuracy	Will not increase prediction accuracy	Will collect sensitive data or violate privacy	Will cause a discriminatory decision	Will cause a biased decision	Other motives	Total number of responses [b]	Total positive motivations	Total negative motivations
Demographic	Age	0.34	0.07	0.19	0.12	0.19	0.09	444	0.34	0.57
	Gender	0.17	0.25	0.12	0.15	0.23	0.08	450	0.17	0.75
	MarStat	0.17	0.28	0.14	0.10	0.19	0.13	432	0.17	0.71
	NoChild	0.16	0.26	0.19	0.07	0.19	0.13	414	0.16	0.71
	IncRate	0.32	0.20	0.12	0.03	0.21	0.12	396	0.32	0.56
Academic	SciSpec	0.52	0.15	0.00	0.10	0.13	0.10	360	0.52	0.38
	SciDeg	0.56	0.08	0.03	0.11	0.15	0.06	372	0.56	0.37
	GPA	0.39	0.23	0.07	0.07	0.13	0.11	366	0.39	0.49
	EduInsEval	0.33	0.23	0.03	0.08	0.26	0.08	396	0.33	0.59
	DegCountry	0.44	0.14	0.03	0.05	0.27	0.08	384	0.44	0.48
Work-related	PrevPerf	0.50	0.10	0.06	0.06	0.13	0.15	372	0.50	0.35
	YrsOrgServ	0.56	0.08	0.03	0.08	0.11	0.13	372	0.56	0.31
	YrsFieldServ	0.65	0.11	0.02	0.07	0.07	0.09	342	0.65	0.26
	NoOrgsWorked	0.53	0.18	0.04	0.07	0.09	0.11	342	0.53	0.37
	TotWorkExp	0.61	0.09	0.04	0.09	0.11	0.07	342	0.61	0.32
	JobTitle	0.39	0.17	0.07	0.08	0.15	0.14	354	0.39	0.47
	JobRank	0.42	0.18	0.02	0.07	0.13	0.18	360	0.42	0.40
	WorkEnvComfort	0.75	0.04	0.04	0.04	0.02	0.12	312	0.75	0.13
	SalarySat	0.71	0.04	0.06	0.04	0.02	0.13	312	0.71	0.15
	RankSat	0.70	0.06	0.06	0.06	0.02	0.11	324	0.70	0.19
	FieldSpecMatch	0.72	0.02	0.04	0.09	0.02	0.11	324	0.72	0.17
	LangOrgMatch	0.57	0.08	0.00	0.15	0.07	0.13	366	0.57	0.30
	TechUseAtt	0.75	0.06	0.00	0.08	0.02	0.09	318	0.75	0.15
Cognitive abilities and personality traits	NumCogTest	0.45	0.23	0.07	0.07	0.04	0.14	336	0.45	0.41
	VerbCogTest	0.42	0.25	0.02	0.09	0.07	0.16	342	0.42	0.42
	LogReasonTest	0.47	0.24	0.00	0.07	0.04	0.18	330	0.47	0.35
	PersTraitTest	0.50	0.21	0.00	0.07	0.05	0.16	336	0.50	0.34
Professional development and certification	TrainHrs	0.67	0.11	0.02	0.06	0.02	0.13	324	0.67	0.20
	WebTrainTime	0.64	0.11	0.04	0.05	0.00	0.16	330	0.64	0.20
	ProfCerts	0.64	0.11	0.02	0.09	0.02	0.13	330	0.64	0.24
	CertBodyEval	0.60	0.07	0.03	0.09	0.05	0.16	348	0.60	0.24

[a]Sample, $n = 306$.
[b]A participant can select one or more motivations that impact their fairness perceptions of each feature.

On the other hand, features such as marital status (MarStat) and number of children (NoChild) raised doubts regarding their role in enhancing prediction accuracy. These personal demographic attributes may be perceived as unrelated to professional capabilities, leading to concerns about their fairness in performance predictions. Similarly, gender is a contentious feature; although it is not inherently indicative of job performance, it seems to raise questions about its relevance to actual work capabilities and its influence on the accuracy of performance prediction.

Concerns about privacy, bias and/or discrimination were evident, particularly with demographic features and other features such as evaluation of the educational institution (EduInsEval) and country from which the degree was obtained (DegCountry). These apprehensions moderated employees' endorsement of such features, revealing that such concerns weigh in their assessment of fairness.

These overall results imply that, while employees are open to incorporating a broad range of features into machine learning models for performance prediction, the emphasis on fairness is critical.

In the following sections, we analyze and discuss the perceptions and motivations associated with each feature across the different dimensions of demographic, academic, work-related, cognitive abilities and personality traits, and professional development and certification.

4.1.1. Demographic-related features

The analysis revealed a considerable level of discomfort with a set of demographic features as predictors of performance. Gender (51% total disagreement), marital status (53%), and number of children (57%) demonstrated a considerable level of disagreement (Figure 1), underpinned by motivations related to bias and discrimination concerns, and privacy violations (Figure 2). These results align with the principle that sensitive or protected attributes should be cautiously considered, if not entirely excluded, from predictive models in machine learning to prevent inherent biases and uphold fairness (Morse et al., 2021). While certain proposed machine learning models incorporate these sensitive attributes, our study underscores the need for heightened awareness of employees' perceptions, advocating for a more ethical and inclusive approach in the utilization of such features for predicting job performance. Furthermore, around a quarter of the surveyed employees did not perceive demographic features (gender, marital status, and number of children) as contributing to the accuracy of performance predictions, further fueling their reluctance to endorse these factors.

The perceptions surrounding the use of "age" as a predictive feature of employee performance present a complex picture. While there is a moderate acceptance of age as a factor (with 55% total agreement), it is notable that a proportion of employees (25% total disagreement) expressed reservations. By deeply investigating the motivations behind these perceptions, we found that approximately one-third of employees believed that including age can enhance the accuracy of performance predictions. However,

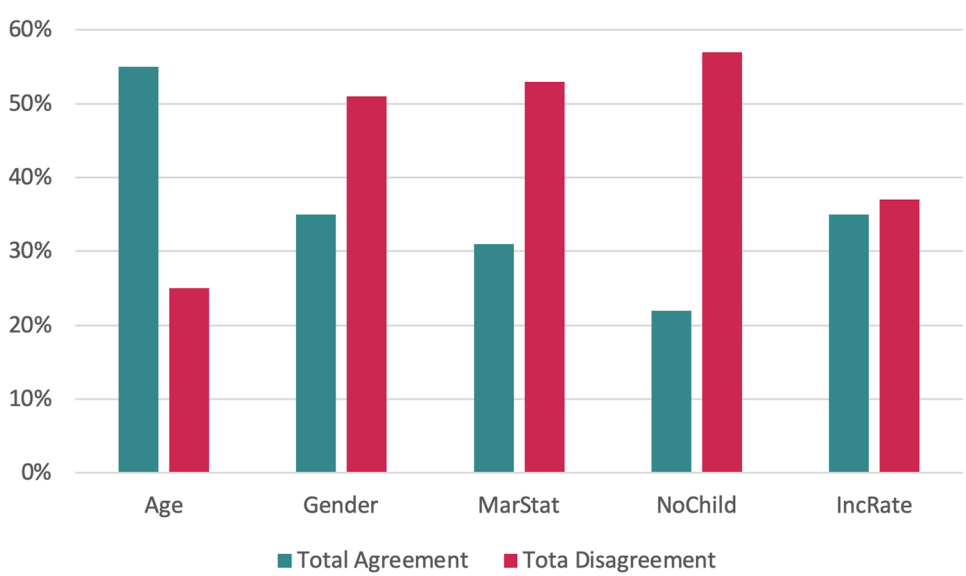

Figure 1. Proportions of employees' perceptions of fairness of the demographic-related features.

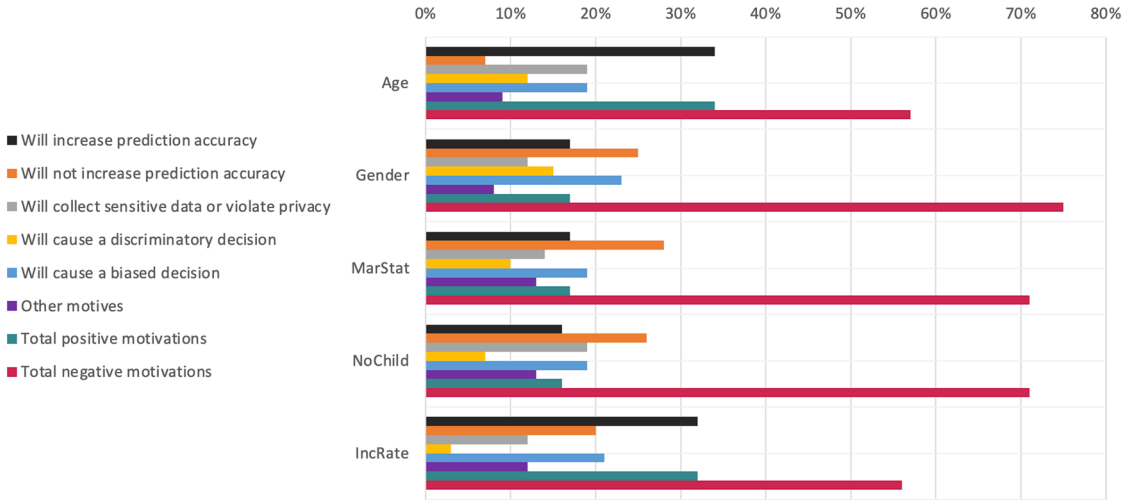

Figure 2. Proportions of motivations of fairness perceptions of the demographic-related features.

more than half, as revealed by the total number of negative motivations, expressed concerns that the use of age might lead to bias, discriminatory decisions, or privacy violations. This dichotomy is further illustrated when examining specific patterns in employee responses. We observed that some employees who acknowledge the potential of age to increase prediction accuracy also had concerns regarding its implications. They might recognize that age is relevant, but its inclusion could lead to privacy issues or biased and discriminatory decisions. This dual perspective could indicate a deep understanding among employees. They are cognizant of the potential benefits of using age as a predictor, yet they are simultaneously aware and cautious of ethical and social ramifications. Such findings underscore the need for a balanced approach to the use of demographic attributes, such as age, in performance prediction models. It highlights the importance of not only considering the potential utility of such features, but also thoroughly addressing and mitigating the associated ethical concerns to maintain fairness and privacy in employee performance evaluations.

The examination of the income rate (IncRate) as a demographic factor in our study revealed a divided stance among employees. The distribution of opinions was relatively balanced, with 35% of employees viewing the income rate as a fair indicator, closely countered by 37% who disagreed with its fairness in performance prediction. Analyzing the motivations behind these perceptions, it became apparent that while a segment of employees (32%) perceived the inclusion of the income rate as potentially enhancing the accuracy of performance predictions, a greater proportion (56%) had apprehensions. These concerns are primarily centered around potential income-based bias and doubts regarding the actual contribution of the income rate to the accuracy of performance prediction. This split in perceptions underscores the complexity in assessing the fairness of the income rate as a predictive feature, reflecting the different perspectives of employees in evaluating the role and impact of this demographic factor in AI-based performance assessment models.

4.1.2. Academic-related features

When considering the perceptions of fairness and the underlying motivations regarding the academic-related features from Figures 3 and 4, we gain comprehensive insight into employee sentiments towards these features. Regarding scientific specialization (SciSpec), the consensus reflects a strong inclination towards acceptance, with 63% of employees perceiving it as a fair predictor of job performance. This perception is buoyed by the positive motivations of 52% of the respondents who advocate for SciSpec's role in enhancing predictive accuracy, suggesting a substantial acknowledgment of the type of specialized knowledge's relevance. For the scientific degrees (SciDeg), we observed an even more pronounced agreement, reaching 71%. This confidence in the fairness of SciDeg is underpinned by 56% of the employees, which is attributed to its capacity to augment prediction accuracy. Despite this, there remains an awareness of the potential biases and other issues associated with the usage of scientific

specialization and degree features in prediction models, as evidenced by 38% and 37% of the negative motivations expressed, respectively.

The grade point average (GPA) yielded a moderate level of agreement (47%) among employees for its use as a fair predictive feature in performance evaluations. While 39% of employees perceived its incorporation as enhancing the accuracy of predictions, a slightly higher proportion (49%) expressed reservations. These concerns were primarily related to their actual impact on prediction accuracy, potential for bias, and privacy issues. This mixed response suggests different views of GPA as a fair predictor.

The evaluation of educational institution (EduInsEval) also evoked a split response, with 47% agreement and 39% disagreement. The results of the underlying motivations for these perceptions showed that although 33% believed in its positive impact on accuracy, 59% of negative motivations pointed towards concerns about using institutional reputation as a predictor. This split indicates the contentious nature of considering educational institution evaluations in performance-prediction algorithms.

Similarly, the country from which the degree was obtained (DegCountry) experienced a moderate level of agreement at 45%, alongside 31% disagreement. While 44% of employees viewed the country of a degree as a factor that could enhance prediction accuracy, negative motivations stood at 48%. This response reflects the complexities and potential controversies in considering the degree's country of origin as a fair and effective predictive feature for employee performance.

Figure 3. Proportions of employees' perceptions of fairness of the academic-related features.

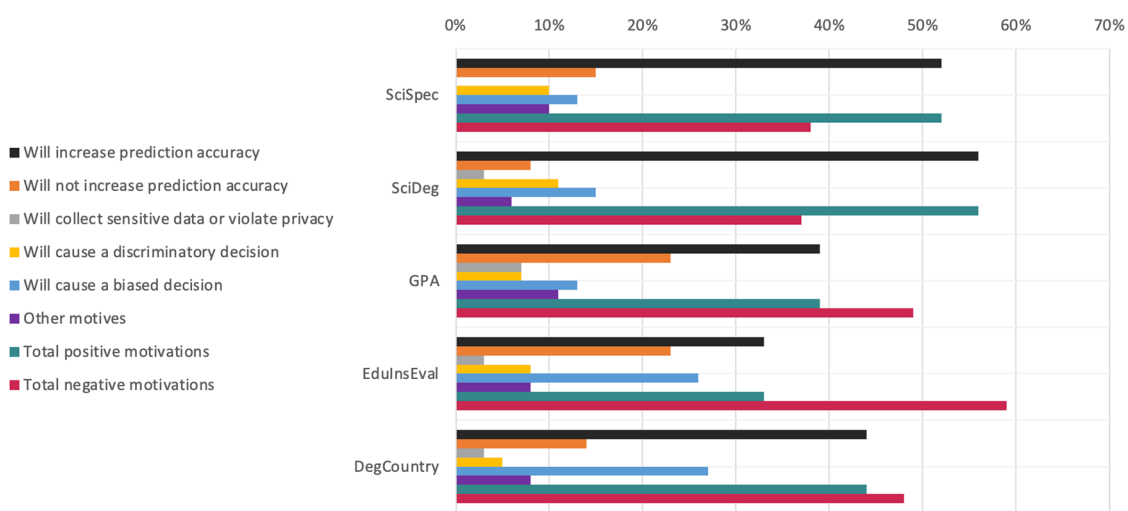

Figure 4. Proportions of motivations of fairness perceptions of the academic-related features.

These results suggest that while academic credentials are valued, there is considerable division over the fairness and relevance of specific aspects, such as GPA, institutional reputation, and country of degree, in AI-driven performance evaluations.

4.1.3. Work-related features

The analysis of employee perceptions concerning the fairness of work-related features, augmented by the motivations underpinning these perceptions, yielded insightful results, as per Figures 5 and 6. A salient feature in this regard is the employee's previous performance (PrevPerf), which received an endorsement as a fair predictor of job performance, as evidenced by 67% of employees concurring with this stance. This agreement was underpinned by a motivational factor, with half of the respondents affirming that PrevPerf can enhance the accuracy of performance predictions.

Employees also considered the years of service in the organization (YrsOrgServ) in high regard as a fair predictor, as evidenced by a 73% agreement rate. This suggests that longevity within an organization is believed to enhance employee performance. From a motivational standpoint, 56% of employees advocated for YrsOrgServ as a contributor to accuracy in performance prediction. However, some concerns remain related to potential bias in decisions that could affect employees with less longevity within an organization.

The years of service in the same field (YrsFieldServ) received a high agreement at 80% as a fair predictor, indicating that field-specific experience is valued for its contribution to performance. Motivationally, 65% affirmed its positive impact on prediction accuracy, reinforcing the premium placed on this predictor.

The number of organizations worked for (NoOrgsWorked) was regarded as a fair measure by 67% of the employees, possibly suggesting a value placed on the breadth of experience as a result of exposure to various organizational cultures and practices. Despite this, the metric is subject to debate, with 53% supporting its role in enhancing prediction accuracy and 37% expressing concerns. The concerns particularly pertained to its actual contribution to prediction accuracy and the potential for biased decisions against employees who have worked for a smaller number of organizations.

Total work experience (TotWorkExp) was viewed positively, with 80% agreement, underscoring a comprehensive appreciation for accumulated professional experience as a fair predictor. Motivationally, 61% perceived it as enhancing prediction accuracy, emphasizing a widespread respect for career longevity.

The agreement rate for job title (JobTitle) was 51%, indicating a moderate belief in its fairness as a measure. However, only 39% of employees considered it to increase prediction accuracy, while 47% harbored reservations about its actual contribution to prediction accuracy, the potential for biased decisions,

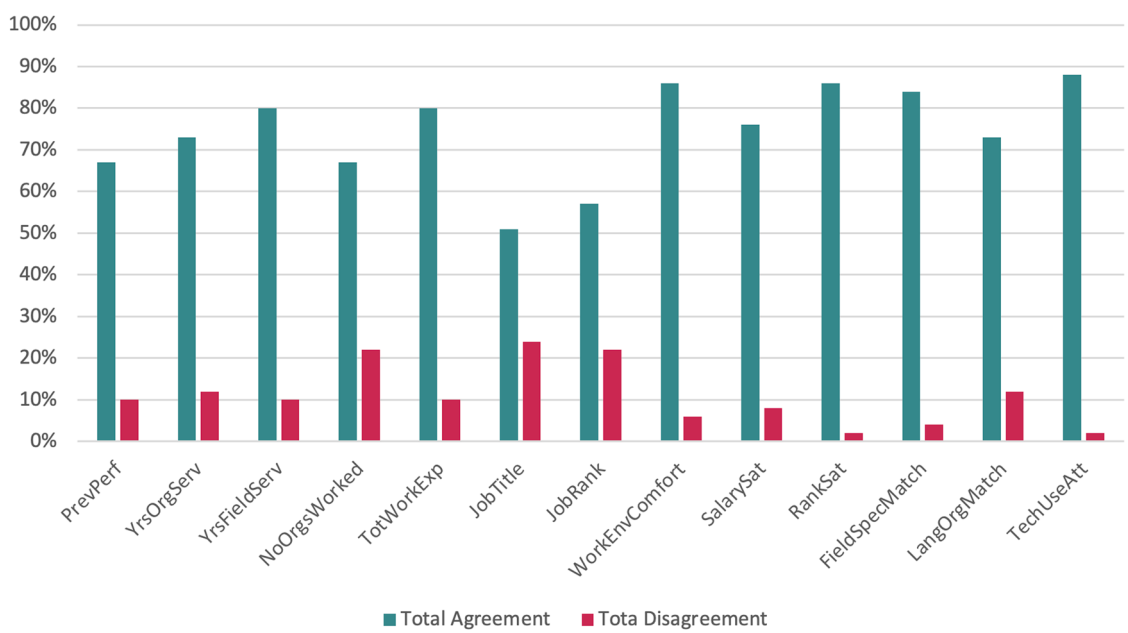

Figure 5. Proportions of employees' perceptions of fairness of the work-related features.

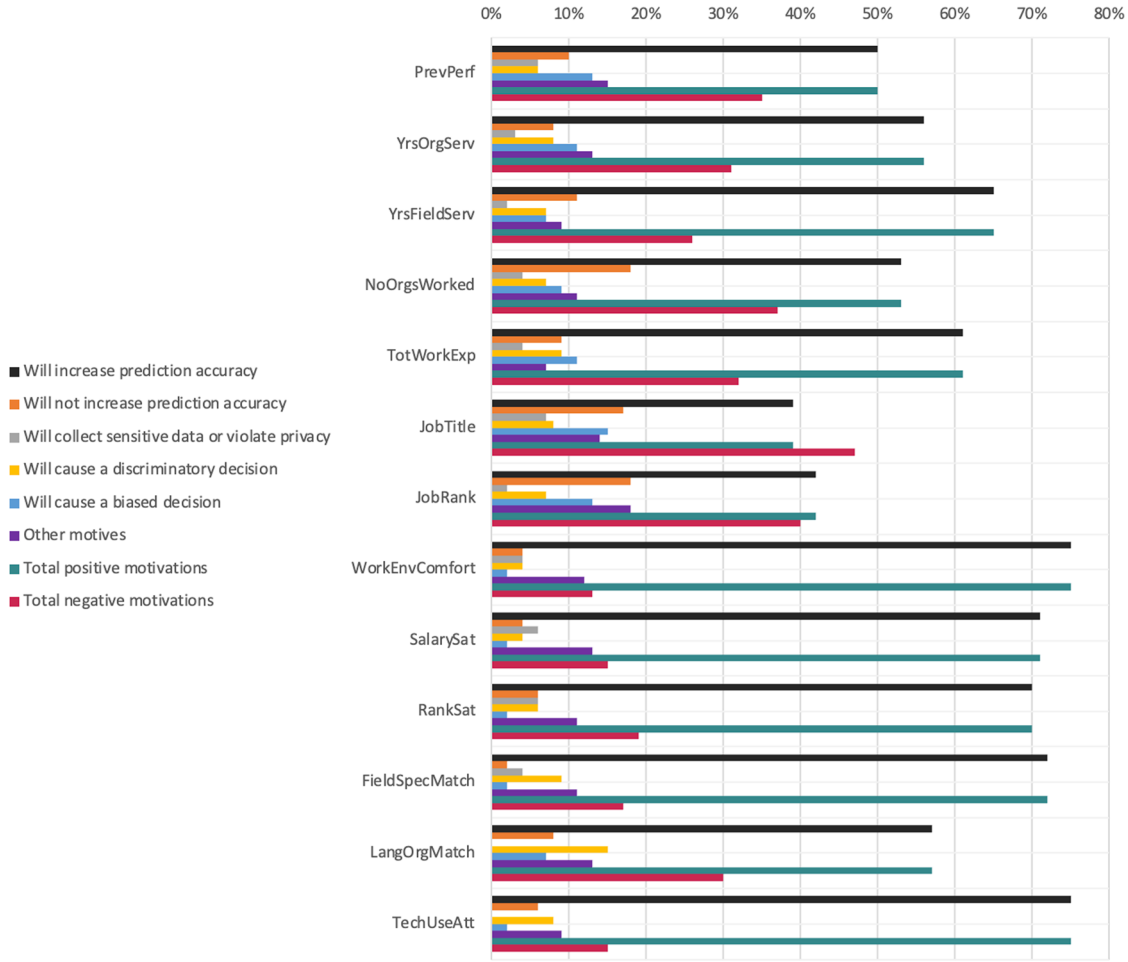

Figure 6. Proportions of motivations of fairness perceptions of the work-related features.

and other issues. Compared to job title, job rank (JobRank) achieved a slightly higher agreement rate of 57%, suggesting that some employees view their hierarchical position as somewhat indicative of their performance capability and, in turn, as a fair measure. Despite this, only 42% regarded it as a relevant measure for enhancing prediction accuracy, and a similar proportion (40%) expressed concerns over its relevance in increasing accuracy, the potential risk of rank-based biased decisions, and other issues.

Work environment comfort (WorkEnvComfort) received a high agreement rate, with 86% of the surveyed employees perceiving it as a fair predictor of performance, suggesting that a positive work environment is crucial for employee performance. A commanding 75% believed that it directly contributes to prediction accuracy, reflecting the high value placed on workplace comfort. Salary satisfaction (SalarySat) is also perceived as a fair performance indicator by 76% of the employees, who likely associate financial recognition with job performance. Motivationally, 71% perceived it as a factor that could enhance prediction accuracy, suggesting that salary satisfaction is closely tied to perceived employee productivity. Rank satisfaction (RankSat) received an 86% agreement rate, indicating that satisfaction with one's job rank within an organization is broadly viewed as a fair predictor of performance. In this case, 70% supported its role in enhancing prediction accuracy.

Field specialization match (FieldSpecMatch) received an 84% agreement rate, indicating that employees widely regard the alignment between their specialized field and their actual work as a fair predictor of performance. A significant 72% of respondents were motivated by the belief that such congruence positively affects prediction accuracy.

The recognition of an employee's native language alignment with the organization's primary communication language (LangOrgMatch) was viewed as a fair indicator of performance, as evidenced by the 73% agreement rate among employees. This consensus rate reflects the acknowledgment of effective communication as one of the cornerstones of workplace productivity. From a motivational perspective,

57% of employees regarded language congruence as bolstering the accuracy of performance prediction. Nevertheless, it is essential to note that, alongside these positive perceptions, there is a thread of concern regarding potential discriminatory practices. A fraction of the workforce expressed apprehension that this criterion may inadvertently disadvantage individuals from diverse linguistic backgrounds, potentially fostering discriminatory decisions against employees of certain nationalities. Such concerns call for a sensitive and inclusive approach to the utilization of LangOrgMatch within performance-prediction models.

Technology usage attitude (TechUseAtt) received the highest consensus, with 88% of the employees agreeing that it serves as a fair measure, mirroring the modern workplace's focus on technological proficiency. An overwhelming 75% of employees consider it valuable for prediction accuracy, highlighting the pivotal role of tech-savviness in today's work environment.

The analysis of work-related features revealed that employees generally regard certain features as fair predictors for performance evaluation. Features such as work environment comfort and field specialization match stood out for their positive impact on prediction accuracy as perceived by employees. However, alongside this endorsement was a consciousness of potential biased or discriminatory decisions, particularly in features such as the match between an employee's native language and the organization's primary communication language (LangOrgMatch). This aspect underscores the delicate balance required to utilize the performance-predictive features within AI models. While these features are valued for their ability to enhance prediction accuracy, as per employees' perceptions, there is obvious awareness among employees about the risks they pose in terms of fairness and ethical implications.

4.1.4. Cognitive abilities and personality traits' features

The results in Figures 7 and 8 concerning cognitive abilities and personality traits' features demonstrate employees' complex views on their fairness in performance prediction. Numerical and verbal cognitive tests (NumCogTest and VerbCogTest) were regarded as fair by 53% and 41% of employees, respectively, albeit with motivations for their fairness being a mix of positive and negative sentiments. This pattern is similarly seen with logical reasoning test (LogReasonTest) and personality trait test (PersTraitTest), receiving fairness agreement rates of 43% and 51%, respectively. Across all these features, there was a notable level of concern about their effectiveness in enhancing performance prediction accuracy, with skepticism ranging from 21% to 25%.

Cognitive and personality tests' features might offer valuable predictive power in machine learning models for employee performance; however, employee perceptions of their fairness must be considered. The mixed perceptions of fairness and the divided motivations highlight the need for transparency and care in the application of these features, ensuring that they serve to enhance the predictive model without compromising fairness.

4.1.5. Professional development and certification features

The perceptions of fairness regarding professional development and certification features revealed a substantial endorsement of these factors as fair indicators for performance prediction (see Figure 9). The total number of training hours (TrainHrs) was regarded as a fair measure, as perceived by 71% of employees, which suggests that instructor-led training is valued for its possible contribution to employee performance. The motivation data showed that 67% of the employees acknowledged that TrainHrs may enhance the accuracy of performance predictions (see Figure 10).

Similarly, time spent on web-based training (WebTrainTime) was seen as a fair indicator by 69% of the employees. This reflects recognition of the importance of self-paced learning and its possible positive impact on job performance. About 64% of respondents believed that WebTrainTime positively influences prediction accuracy, underscoring the perceived direct link between ongoing learning and job performance.

Professional certifications (ProfCerts) was also seen in a favorable light, with a 73% agreement rate as a fair measure. The acquisition of professional certifications was considered by 64% of employees to boost the accuracy of performance predictions, highlighting the value placed on the formal recognition of skills and competencies as a fair performance predictor.

Figure 7. Proportions of employees' perceptions of fairness of the cognitive abilities and personality traits' features.

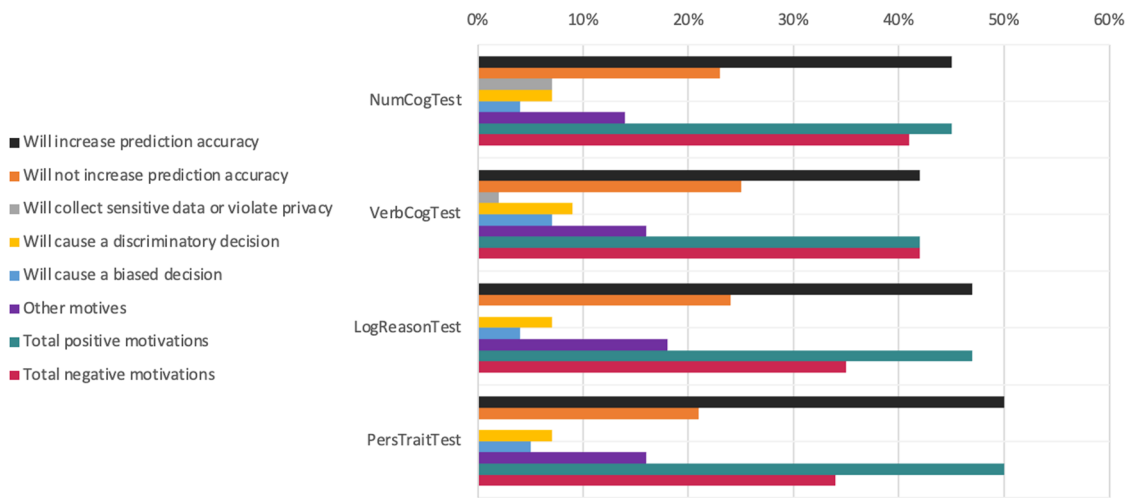

Figure 8. Proportions of motivations of fairness perceptions of the cognitive abilities and personality traits' features.

The evaluation of professional certification bodies (CertBodyEval) was viewed as a fair measure by 73% of the employees, indicating that the credibility and standing of certification bodies play a role in the perceived value of certifications. Sixty percent of the surveyed employees supported the notion that CertBodyEval is beneficial for enhancing prediction accuracy, suggesting trust in this feature as a relevant predictor.

When synthesized, these results suggest a strong belief among employees regarding the positive impact of professional development and certificates on performance. There is a clear implication for machine learning models used for performance prediction; that is, incorporating metrics related to training and certifications might improve the models' accuracy, as perceived by employees. However, it is crucial to ensure that such features are appropriately weighed, considering the variety of professional paths and learning styles. Over-reliance on formal certifications without acknowledging experiential learning could introduce unfairness and overlook the full range of employee capabilities. As organizations seek to harness the predictive power of machine learning, these insights emphasize the importance of a balanced and comprehensive approach that values both the formal and informal aspects of professional development.

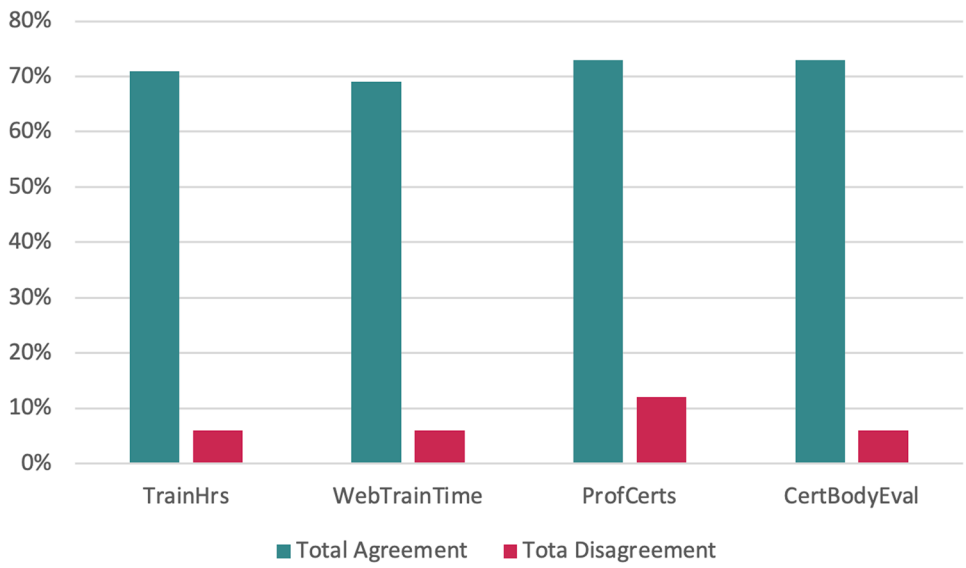

Figure 9. Proportions of employees' perceptions of fairness of the professional development and certification features.

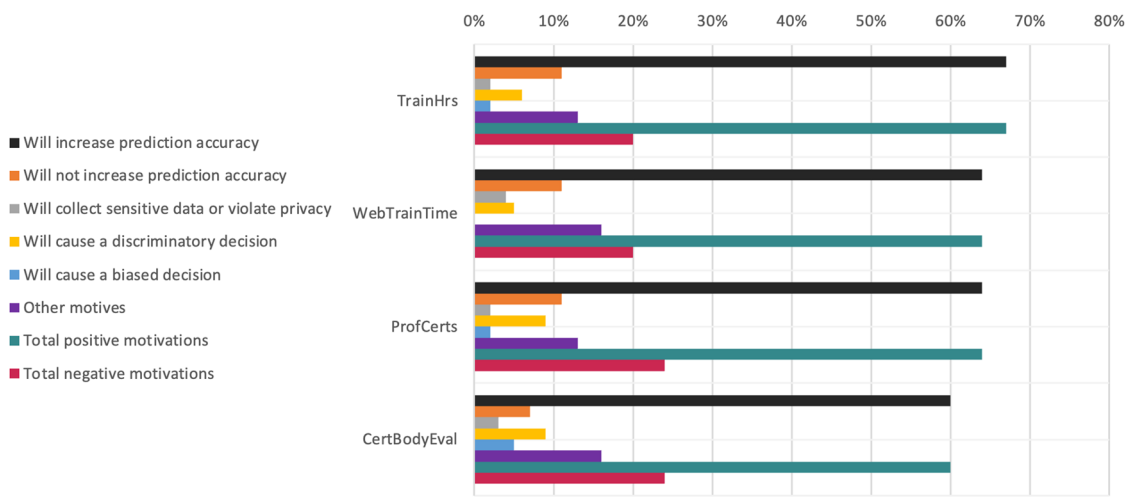

Figure 10. Proportions of motivations of fairness perceptions of the professional development and certification features.

4.2. Correlation matrix

We observed correlations between fairness perceptions of many predictive features of employee performance. Figure 11 shows the correlation matrix, which shows how the fairness perceptions of features correlate with each other. For example, there is a strong positive correlation (0.78) between gender and marital status (MarStat), suggesting a parallel perception of the fairness of these two features in the context of AI-based employee performance prediction. This means that employees who deem gender a fair criterion for predicting performance are inclined to view marriage status as fair. Conversely, if they regard gender as an unfair basis for performance prediction, they are likely to consider marital status unfair. This correlation indicates that these demographic factors are interconnected in employees' judgments about the fairness of AI performance predictions. Similarly, the strong positive correlations between the number of children (NoChild) and gender (0.78), as well as between the number of children and marital status (0.81), further reinforce this pattern. Employees who perceive the number of children as a fair or unfair basis for performance prediction also tend to have similar perceptions of fairness or unfairness regarding both gender and marital status. These interconnected views highlight employees' internal consistent perceptions when evaluating some demographic factors as fair or unfair features for AI-based performance assessments.

Similarly, the strong positive correlation of 0.76 between scientific degree (SciDeg) and scientific specialization (SciSpec) suggests that employees often view these academic factors in tandem when considering fairness in AI-based performance prediction. This correlation indicates that if an employee believes that a scientific degree is a fair basis for performance evaluation by AI, they are also likely to consider the field of scientific specialization as a fair criterion. In the same context, the strong positive correlation (0.83) between the country of degree (DegCountry) and the evaluation of the educational institution (EduInsEval) further emphasizes this pattern. Employees who place importance on the country in which the degree was obtained for fair AI performance assessment are also inclined to regard the reputation or evaluation of the educational institution as an important factor. These correlations suggest a holistic approach by employees in considering specific academic features as fair or unfair measures in AI-based performance predictions.

In the context of workplace experience, a strong positive correlation of 0.75 between years of service in a particular field (YrsFieldServ) and years of service within the same organization (YrsOrgServ) indicates that employees commonly view these two aspects similarly when assessing fairness in AI-based performance prediction. Similarly, a robust positive correlation of 0.72 exists between the number of organizations an employee has worked for (NoOrgsWorked) and their years of service in the current organization (YrsOrgServ). This correlation implies that employees who view one of these features as a fair or unfair basis for performance prediction are likely to perceive the other in the same manner.

Furthermore, the strong correlations of 0.69 and 0.67 between total work experience (TotWorkExp) and years of service in the same field (YrsFieldServ) and organization (YrsOrgServ) respectively, suggest that employees perceive their overall work history, including all past professional experiences, as

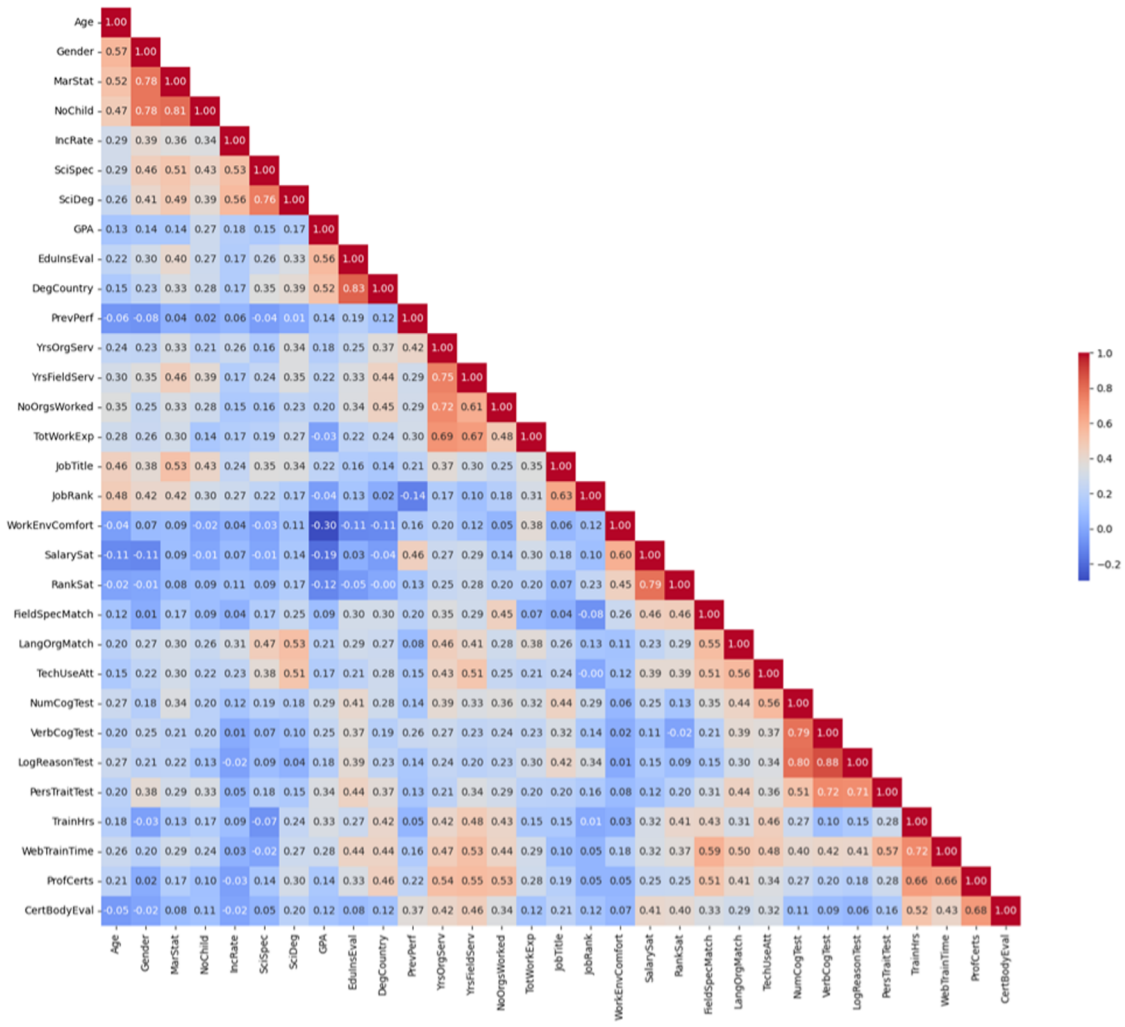

Figure 11. Correlation matrix.

inherently connected to their tenure in a specific field and organization. This perspective on the fairness of the AI-driven performance assessment context indicates a consistent approach among employees, indicating that they generally assess the fairness of work experience features similarly. This uniformity in perception highlights the collective value that employees place on both the breadth and depth of professional experience when considering the fairness of work experience features in performance evaluations.

Given the strong positive correlation of 0.79 between salary satisfaction (SalarySat) and job rank satisfaction (RankSat), it can be inferred that employees' perceptions of fairness of these two features are closely aligned. This means that when employees perceive the satisfaction with salary as a fair (or unfair) criterion for AI-based performance evaluation, they tend to perceive the satisfaction with job rank as fair (or unfair). The strength of this correlation highlights a significant link in the way employees evaluate the fairness of these two aspects of their employment in the context of AI-based job performance assessments.

Within the domain of cognitive and personality assessments, the correlations revealed insightful patterns of employees' perceptions of fairness in the AI-based performance evaluation. A strong positive correlation (0.79) between the verbal and numerical cognitive tests (VerbCogTest and NumCogTest) indicates that employees who perceive the verbal cognitive test as fair for performance prediction are also likely to regard the numerical cognitive test as fair. Similarly, a strong positive correlation of 0.80 between logical reasoning and numerical cognitive tests (LogReasonTest and NumCogTest) suggests a shared perspective among employees regarding the fairness of these cognitive assessments in predicting performance. Similarly, the strong positive correlation of 0.88 between the logical reasoning test (LogReasonTest) and verbal cognitive test (VerbCogTest) demonstrates that employees who value logical reasoning as a fair measure often regard verbal cognitive abilities as a fair feature. Furthermore, the strong correlations of 0.72, and 0.71 between personality trait test (PersTraitTest), and verbal cognitive and logical reasoning tests (VerbCogTest and LogReasonTest) respectively implies that employees who consider personality assessments as fair are inclined to also view verbal cognitive and logical reasoning abilities as fair predictors. Collectively, these correlations point to a holistic view among employees, where a range of cognitive abilities and personality traits are considered interconnected in the context of fair AI-driven performance predictions.

In the area of professional development and certification, the observed correlations suggest a cohesive view among employees regarding the fairness of specific factors in AI-based performance predictions. A strong positive correlation (0.72) between the total number of instructor-led training hours (TrainHrs) and the time spent on self-paced web-based training (WebTrainTime) indicates that employees who consider the number of training hours as a fair metric for performance assessment are also likely to value the time dedicated to web-based training in the same manner. This reflects a similar fairness perception of both instructor-led training hours and self-paced online training engagement time. Additionally, a strong positive correlation of 0.66 between the number of professional certificates (ProfCerts) and training hours (TrainHrs) suggests that employees who perceive the number of professional certifications as a fair predictor also consider the total hours spent in training as a fair indicator for performance prediction. Moreover, the correlation of 0.66 between professional certificates (ProfCerts) and time spent on web-based training (WebTrainTime) indicates that employees who perceive number of professional certifications as a fair feature also tend to perceive the time invested in web-based training as a fair predictor. This indicates a shared perception of both formal certification and web-based learning in the context of skill development as (un)fair features. Finally, the correlation of 0.68 between the evaluation of professional certification bodies (CertBodyEval) and the number of professional certificates (ProfCerts) underscores the importance that employees place on the reputation and quality of certifying bodies, in conjunction with the certificates obtained when perceiving the fairness of using them as performance prediction features.

5. Conclusion and future works

This study investigated 306 employees' perceptions of the fairness of 31 features that might be used in AI-based job performance assessment systems. Features related to attitude towards technology use, workplace comfort, job rank satisfaction, and alignment between one's field of expertise and

work received a strong consensus on their fairness and relevance. However, the study also revealed critical concerns regarding the incorporation of certain demographic attributes, with implications for privacy, potential biases, and discrimination. These reservations highlight the need for a careful and ethically informed approach to the selection of performance predictors in AI models, ensuring that they are not only effective, but also do not violate privacy or cause unfair, discriminatory, or biased decisions.

The study's findings, illustrated by strong correlations among a set of features, emphasize the interconnected nature of fairness perceptions in AI-based employee performance evaluations. These insights underscore the necessity of a holistic approach in the development and implementation of AI algorithms to ensure that they align with the multifaceted criteria employees deem fair and equitable.

The limitations of this study offer opportunities for further research. The relatively small sample size highlights the need for a more extensive and diverse dataset that allows the investigation of the interplay between demographic characteristics and perceptions. Therefore, future studies can be conducted to advance our understanding of how individual differences and backgrounds shape fairness perceptions.

The participants in our study were all from Saudi Arabia, which limits the generalizability of the findings. Future research should include participants from various countries to provide a more comprehensive understanding of fairness perceptions in different cultural contexts. However, despite this limitation, our study provides a significant contribution by shedding light on the perspectives of Saudi Arabian employees, a demographic often underrepresented in AI fairness-related research. This focus could offer unique findings and insights into the perceptions of performance-predictive features in AI algorithms within a non-western context.

There is also the opportunity for future research to investigate the development of culturally aware and fair predictive algorithms. This involves leveraging the findings and insights from our study and comparing them with findings from similar research conducted in diverse cultural settings. The aim is to design AI job performance prediction models that not only align with global fairness standards, but also respect and integrate cultural preferences.

Finally, longitudinal studies can also provide insights into how fairness perceptions of performance-predictive features in AI algorithms evolve over time, particularly as employees witness the practical outcomes of AI-based assessments.

The practical implications of this study underscore the critical importance of adopting an ethically informed approach to the development and implementation of AI-based job performance assessment systems. By prioritizing predictive features directly related to job performance, such as attitude towards technology use, workplace comfort, job rank satisfaction, and alignment between expertise and work, organizations might enhance the perceived fairness and effectiveness of these systems. However, caution must be exercised in incorporating demographic attributes, such as gender or marital status, to avoid privacy breaches and biased decisions. To mitigate the risk of bias, organizations should either exclude sensitive demographic features or apply fairness-aware algorithms that adjust for potential biases in the prediction process.

To enhance transparency, organizations should ensure that employees understand how AI models are used in performance assessments by explaining which features are included and how they influence decisions. This transparency can be supported through regular briefings and AI literacy training, helping employees build trust in the system. Furthermore, developing ethical guidelines for AI implementation, in collaboration with AI ethicists and HR professionals, will ensure that performance prediction algorithms are fair, transparent, and aligned with legal standards as well as employee perspectives.

Organizations should also recognize the interconnected nature of fairness perceptions of performance-predictive features in AI algorithms and adopt a holistic approach to algorithm development. Furthermore, efforts to collect diverse and representative datasets, prioritize cultural sensitivity in algorithm design, and conduct longitudinal studies on fairness perceptions can enhance both the practical relevance and ethical integrity of AI-based performance assessments. By implementing these recommendations, organizations may improve their AI-based performance assessment systems, ensuring they meet employee expectations of fairness while mitigating potential biases and discrimination.

Data

Due to ethical and legal restrictions, supporting data is not available.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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